THE SUM OF A PERFECT COMPLEX AND ITS SHIFT ON A QUASI-COMPACT OPEN OF AN AFFINE SCHEME EXTENDS TO A PERFECT COMPLEX

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Allowed standard prerequisites (editorial).

- For a perfect complex F on a quasi-compact open of an affine scheme, the object F plus a shifted finite locally free module is the restriction of a perfect complex on the whole scheme. invoked at proof lines 3622-3626. Provides the initial perfect extension E whose restriction is F[r] plus F.
- Maps of perfect complexes can be lifted along restriction to the open, with prescribed restriction. invoked at proof lines 3627-3638. Produces the endomorphism alpha of the extension acting as the projection onto the locally free summand over U.

Source reference. Original lemma 08EG: sources/stacks-perfect/pinned-original-source.tex, source lines 3525–3525. Cited at proof lines 3622-3626 to obtain E; the construction of alpha and the cone are inside the counted span.

Source reference. Original lemma 08EH: sources/stacks-perfect/pinned-original-source.tex, source lines 3571–3571. Cited at proof lines 3627-3638 to lift the projection endomorphism; the cone construction finishing the proof is inside the counted span.

Lemma 0.2. *Let X be an affine scheme. Let $U \subset X$ be a quasi-compact open. For every perfect object F of $D(\mathcal{O}_U)$ the object $F \oplus F[1]$ is the restriction of a perfect object of $D(\mathcal{O}_X)$.*

Proof. By Lemma 08EG we can find a perfect object E of $D(\mathcal{O}_X)$ such that $E|_U = \mathcal{F}[r] \oplus F$ for some finite locally free $\mathcal{O}_U$-module $\mathcal{F}$. By Lemma 08EH we can find a morphism of perfect complexes $\alpha : E_1 \to E$ such that $(E_1)|_U \cong E|_U$ and such that $\alpha|_U$ is the map

$$\begin{pmatrix} \mathrm{id}_{\mathcal{F}[r]} & 0 \\ 0 & 0 \end{pmatrix} : \mathcal{F}[r] \oplus F \to \mathcal{F}[r] \oplus F$$

Then the cone on α is a solution. □

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